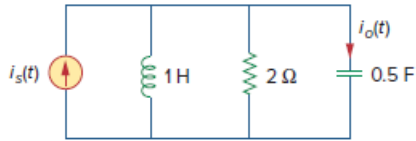


Problem

If $i_s(t) = 7.5e^{-2t}u(t)$ A in the circuit shown in Fig. 16.40, find the value of $i_o(t)$.

Figure 16.40:



Step-by-step solution

Step 1 of 6

The initial current through the inductor $i(0^-) = 0$ and voltage across capacitor is $v_o(0^-) = 0$.

Convert the elements in the circuit from time domain to s – domain.

Apply the Laplace Transform for the following elements in the circuit:

For input current element:

$$L[I_s(t)] = L[7.5e^{-2t}u(t)]$$

$$I_s(s) = 7.5L[e^{-2t}u(t)]$$

$$I_s(s) = \frac{7.5}{s+2}$$

For Inductor element:

The equivalent value in s-domain is:

$$\begin{aligned} L[sL] &= L[s(1)] \\ &= s \end{aligned}$$

For capacitor element is,

$$\begin{aligned} L\left[\frac{1}{sC}\right] &= L\left[\frac{1}{s(0.5)}\right] \\ &= \frac{2}{s} \end{aligned}$$

Step 2 of 6

Draw the s – domain equivalent circuit.

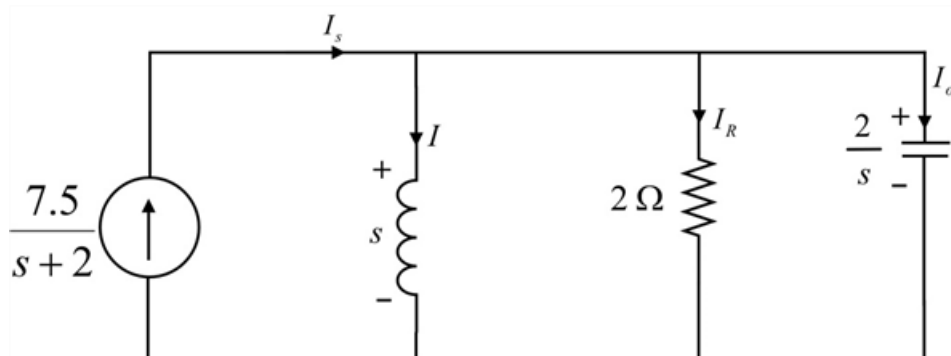


Figure 1

Step 3 of 6

Find the equivalent resistance of the circuit.

Components inductors, capacitors and resistors are in parallel.

$$\frac{1}{Z(s)} = \frac{1}{s} + \frac{1}{2} + \frac{s}{2}$$

$$\frac{1}{Z(s)} = \frac{2+s+s^2}{2s}$$

$$\frac{1}{Z(s)} = \frac{s^2+s+2}{2s}$$

$$Z(s) = \frac{2s}{s^2+s+2}$$

Step 4 of 6

Apply the ohm's law to the circuit at input supply.

$$V(s) = I(s)Z(s)$$

$$= \left(\frac{7.5}{s+2} \right) \left(\frac{2s}{s^2+s+2} \right)$$

$$V(s) = \left(\frac{7.5}{s+2} \right) \left(\frac{2s}{s^2+s+2} \right) \dots\dots (1)$$

Find the current flow through the capacitor using ohm's law.

$$I_0(s) = \frac{V(s)}{\frac{2}{s}}$$

$$= \frac{s}{2} V(s)$$

$$= \left(\frac{s}{2} \right) \left(\frac{7.5}{s+2} \right) \left(\frac{2s}{s^2+s+2} \right)$$

$$= \frac{7.5s^2}{(s+2)(s^2+s+2)}$$

Step 5 of 6

Take the partial fraction expansion to the current expression $I_0(s)$.

$$I_0(s) = \frac{7.5s^2}{(s+2)(s^2+s+2)}$$

$$= \frac{7.5}{s+2} - \frac{7.5}{s^2+s+2}$$

$$= \frac{7.5}{s+2} - \frac{7.5}{(s+0.5)^2 + 1.75}$$

$$= (7.5) \left(\frac{1}{s+2} \right) - \left(\frac{7.5}{\sqrt{1.75}} \right) \left(\frac{(\sqrt{1.75})}{(s+0.5)^2 + (\sqrt{1.75})^2} \right)$$

Step 6 of 6

Apply inverse Laplace transform.

$$\begin{aligned}i_o(t) &= \left(7.5e^{-2t} - \frac{7.5}{\sqrt{1.75}} e^{-0.5t} \sin(\sqrt{1.75}t) \right) u(t) \\&= \left(7.5e^{-2t} - \frac{7.5}{\sqrt{\frac{7}{4}}} e^{-0.5t} \sin\left(\sqrt{\frac{7}{4}}t\right) \right) u(t) \\&= 7.5 \left(e^{-2t} - \frac{2}{\sqrt{7}} e^{-0.5t} \sin\left(\frac{\sqrt{7}}{2}t\right) \right) u(t)\end{aligned}$$

Hence, the current $i_o(t)$ flow through the capacitor is

$$7.5 \left(e^{-2t} - \frac{2}{\sqrt{7}} e^{-0.5t} \sin\left(\frac{\sqrt{7}}{2}t\right) \right) u(t) \text{ A}$$